

Optimising Sales Performance in the Context of CRM

Domma Bat (Dolgormaa Bat) — MSc Business Analytics, Lancaster University Management School

PROGRAMME	PLACEMENT	METHODS	SUBMITTED
MSc Business Analytics	Sustainable Energy First, Manchester	Logistic regression · KNN · Clustering	September 2023

This is a short excerpt — the executive summary and introduction — from a longer dissertation, shared to give a sense of the writing and the analytical approach. Client-specific figures and data have been generalised. The full dissertation and R code are available on request.

Executive Summary

Competition for customers is intense in most commercial markets, and understanding customers well enough to serve them appropriately has become a genuine source of advantage. One route to that understanding is an effective Customer Relationship Management (CRM) system — yet many organisations use CRM technology without extracting much value from it. The gap is usually explained by Critical Success Factors (CSFs): the conditions that determine whether a CRM system actually improves how a business understands and retains its customers. While the specific factors vary by organisation, the research literature points to a recurring set of CSFs and associated metrics.

This dissertation set out to identify which CSFs applied to a business-to-business energy services company, using the company's own contract and customer data, and to build models that put those CSFs to practical use. Working through the available data, three CSFs stood out as most relevant: customer segmentation, customer information management, and customer service.

To address customer segmentation, two clustering approaches — hierarchical clustering and expectation-maximisation (EM) — were applied to the customer base, with a comparative evaluation of the two methods. To address customer information management and customer service, a binary "re-entry" variable was constructed to capture whether a customer returned to the company after a contract ended, used here as an indirect proxy for customer satisfaction in the absence of a formal satisfaction survey. Logistic regression was used to model re-entry, validated against a K-Nearest Neighbours (KNN) classifier as a comparative check. Both classification models performed strongly.

The dissertation concludes that logistic regression and KNN are practical, interpretable tools for modelling customer re-entry, while hierarchical clustering and EM provide complementary, nuanced views for segmentation. Together, the two strands of analysis were designed to feed directly into the company's CRM strategy rather than remain a purely academic exercise.

Chapter 1 — Introduction

A company's relationship with its customers is fundamental to its business success, in much the same way relationships are fundamental to human life more broadly. CRM technology is one of the clearer routes to strengthening that relationship — but its value depends entirely on how well it is used, which is where Critical

Success Factors (CSFs) come in. Improving the metrics behind the relevant CSFs is, in effect, how a company turns CRM from a system it owns into a system that does something for it. This study focuses on optimising sales performance within the CRM function of a business-to-business energy services provider.

The primary objective was to identify the CSFs most likely to make the company's CRM system more effective, and by extension, to improve its sales performance. Drawing on prior research — including studies identifying over fifty CSFs in some contexts, and more focused frameworks proposing a smaller core set with associated metrics — the task was to work out which of these applied to this company, based on the data actually available. Three CSFs emerged as most applicable: customer segmentation, customer information management, and customer service.

Two structured datasets — contract information and customer information — were built from the company's underlying data to support the analysis. The resulting dataset covered a substantial multi-year population of customers and contracts. To manage data quality concerns in the earliest years of the underlying records, and in line with the company's own recommendation, the modelling was restricted to a more recent multi-year window.

Customer information management and customer service were explored through a classification approach: a binary "re-entry" variable, marking whether a customer returned to the company after a contract ended, was used as an indirect measure of satisfaction in the absence of a formal survey. Logistic regression and K-Nearest Neighbours classification were both applied, and both achieved strong predictive performance — giving the company a practical, ongoing way to monitor attrition and understand customer behaviour, independent of this one-off study.

Customer segmentation was explored through clustering. Six variables — spanning sale value, contract duration, number of contracts, share of savings, sector, and contract type — were selected to give a well-rounded view of the customer base. Hierarchical clustering and expectation-maximisation were applied and compared, producing a small number of distinct customer segments, each with a clearly different profile, intended to sharpen the company's engagement strategy.

The scope of the project reflects the data and time window available at the time, but the methods and the way the problem was structured are intended to generalise — the same approach should hold up against later data as the business evolves.